



**Escola Superior
de Tecnologia
e Gestão**

Politécnico de Coimbra

Inteligência Artificial

**Independence Tests
Association Measures
Correlation Measures**

Table of Contents

- **Independence Tests**
 - Pearson's chi-squared test (χ^2 test)
 - Resíduos Standard Ajustados
 - Pressupostos
- **Association Measures**
 - Phi
 - V de Cramer
 - Coeficiente de Contingência
- **Correlation Measures**
 - Spearman Coefficient
 - Pearson Coefficient
- **SPSS**

Inferential Statistics

- **Independence Tests**
- Several test statistics are available for testing the independence of categorical variables from two-way contingency tables:
 - **Pearson's chi-squared test (χ^2 test)**
 - **Likelihood ratio test**
 - **Fisher's exact test**
 - para tabelas 2x2, quando não se pode aplicar o teste χ^2

Inferential Statistics

- Independence Tests
- Pearson's chi-squared test (χ^2 test)
 - determines whether there is an association between categorical variables (i.e., whether the variables are independent or related).
 - it is a nonparametric test.
 - used to determine whether there is a statistically significant difference between the expected frequencies and the observed frequencies in one or more categories of a contingency table.

sig < 0,05 » Rejeita-se a hipótese nula da independência das variáveis

sig, p-value, p:

- se obtivermos sig = 0,000 citar como **sig < 0.001**
- Para obter "sig one-tailed" deve dividir-se "sig 2-tailed" por 2

Inferential Statistics

- Independence Tests
- Pearson's chi-squared test (χ^2 test)
 - Resíduos Standard Ajustados
 - Quanto maior o módulo do resíduo ajustado ($> 1,96$) numa categoria, mais essa categoria se afasta da H_0 de independência:
 - $[-\infty; -1,96]$: associação negativa
 - $[-1,96; 1,96]$: sem associação
 - $[1,96; +\infty]$: associação positiva
 - resulta do cálculo do resíduo em cada célula (valor observado menos o valor esperado) dividido por uma estimativa do erro-padrão sendo expresso em unidades de desvio-padrão acima ou abaixo da média.

Inferential Statistics

- Independence Tests
- Pearson's chi-squared test (χ^2 test)
 - Pressupostos
 - não haver mais do que 20% de células com valor esperado < 5
 - nenhuma célula com valor esperado < 1
 - Para ultrapassar o problema de infração das regras poder-se-á tentar agregar, quando possível e fizer sentido, as classes mais pequenas

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Inferential Statistics

- **Association Measures**
- the **chi-square** test shows if two nominal variables are (in)dependent, but:
 - its value does not evaluate the relative strength of a existing statistically significant relationship, and it is influenced by the sample dimension
 - higher chi-square values do not mean greater relationship between the variables
- **measures of association**
 - help evaluate the relative intensity of a statistically significant relationship.
 - they are not used in interpreting the data unless the chi-square test first shows there is statistical significance (i.e., it doesn't make sense to say there is a strong relationship between two variables when your statistical test shows this relationship is not statistically significant).

Inferential Statistics

- **Association Measures**

Medida de Associação	Aplicação
Phi	Nominal vs. Nominal <i>(ou quando a ordem não for tomada em consideração, e portanto, pode aplicar-se quando temos Ordinal vs. Nominal e Ordinal vs. Ordinal⁶)</i>
V de Cramer	
Coeficiente de Contingência	
Tau B de Kendall Tau C de Kendall	Ordinal vs. Ordinal <i>(ou seja, a ordem é tomada em consideração)</i>
Eta Eta ²	Nominal vs. Intervalo <i>(ou quando uma outra escala for tratada como nominal⁷)</i>

Inferential Statistics

- **Association Measures**

Medida	Cálculo	Valores que assume e comentário
Phi	$\text{Phi} = \sqrt{\frac{\chi^2}{n}}$	Só em tabelas 2x2 é que assume valores entre 0 e 1.
V de Cramer	$V = \sqrt{\frac{\chi^2}{n(q-1)}}$	$0 \leq V \leq 1$ (em tabelas 2x2 o Phi é igual ao V de Cramer)
Coeficiente de Contingência	$C = \sqrt{\frac{\chi^2}{\chi^2 + n}}$	$0 \leq C \leq \sqrt{\frac{q-1}{q}}$, onde $q = \min(r, c)$, ou seja, é o menor número de entre o número de linhas e colunas; portanto, mesmo no caso de associação perfeita nunca assume o valor 1.

Inferential Statistics

- **Association Measures**

- a coefficient of association is independent of its measurement scale. These coefficients typically range between 0 and 1, where 0 is no relationship and 1 is a perfect relationship.

0.9 positivo indica uma associação muito forte.

0.7 a 0.9 positivo indica uma associação forte.

0.5 a 0.7 positivo indica uma associação moderada.

0.3 a 0.5 positivo indica uma associação fraca.

0 a 0.3 positivo ou negativo indica uma associação desprezível.

H_0 = valores não significativos: (sig, p-value, p) < 0,05 » valores significativos

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Inferential Statistics

- **Correlation Measures**

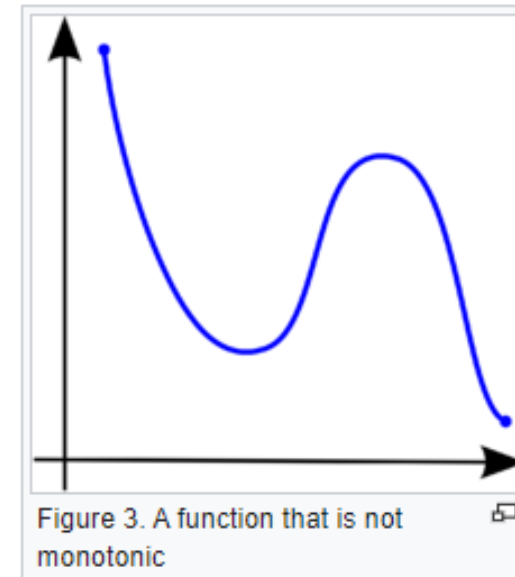
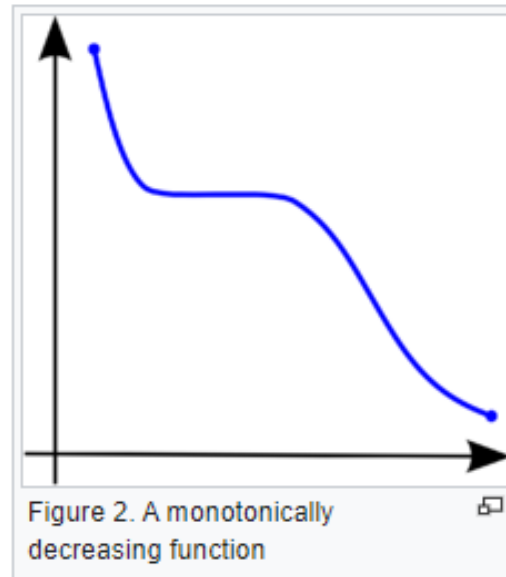
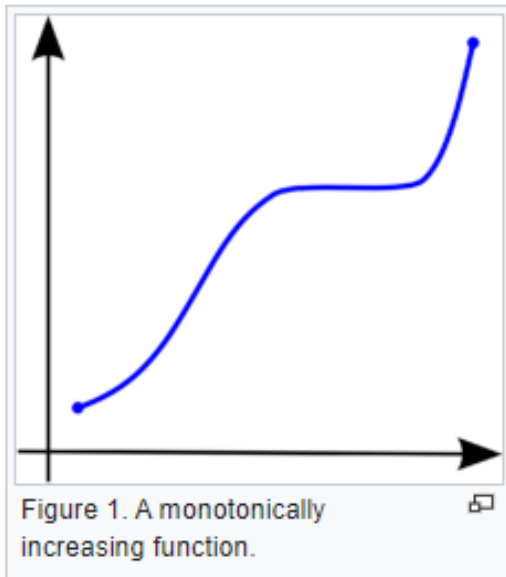
- **correlation** is the degree to which two variables are linearly related.
- the correlation measures give by itself the intensity of the linear association between two variables:
 - **Spearman Correlation Coefficient**
 - **Pearson Correlation Coefficient**
- The values range between -1.0 and 1.0:
 - -1.0 shows a perfect negative correlation
 - 0.0 shows no linear relationship
 - +1.0 shows a perfect positive correlation

Inferential Statistics

- **Correlation Measures**

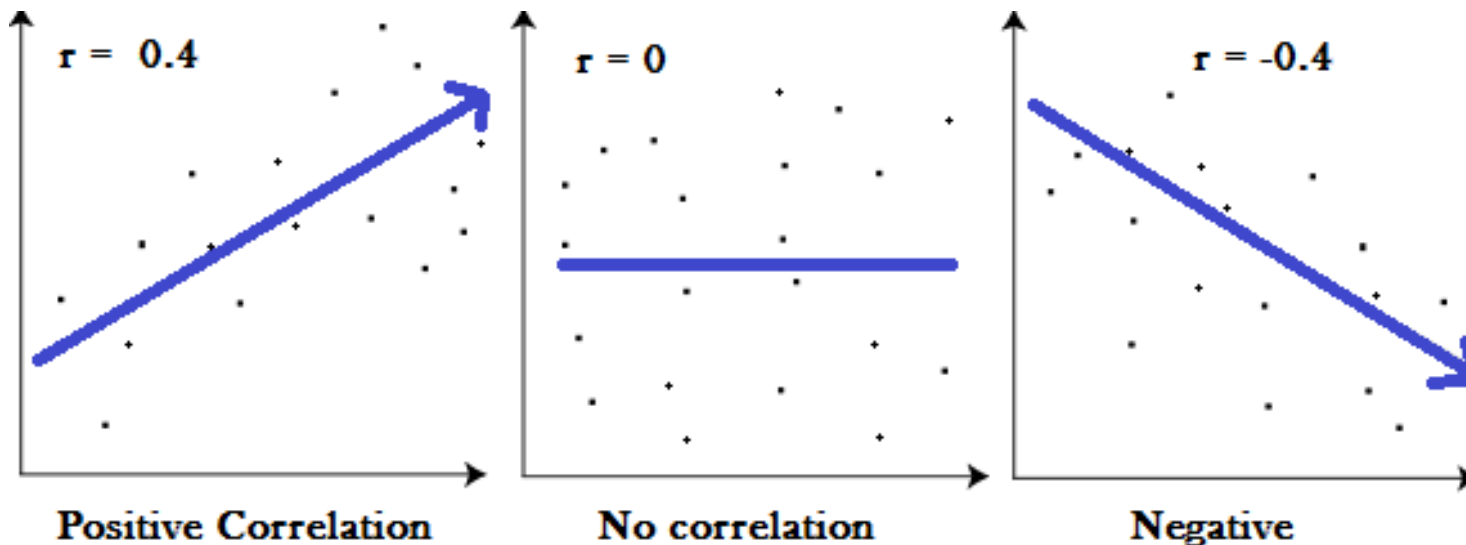
- **Spearman Correlation Coefficient:**

- evaluates a monotonic relationship between two variables - continuous or ordinal
- no restrictions regarding the sample distribution



Inferential Statistics

- **Correlation Measures**
 - **Pearson Correlation Coefficient**
 - quantitative x quantitative
 - requires a normal distribution



Inferential Statistics

- Correlation Measures
 - correlation does not mean causation.

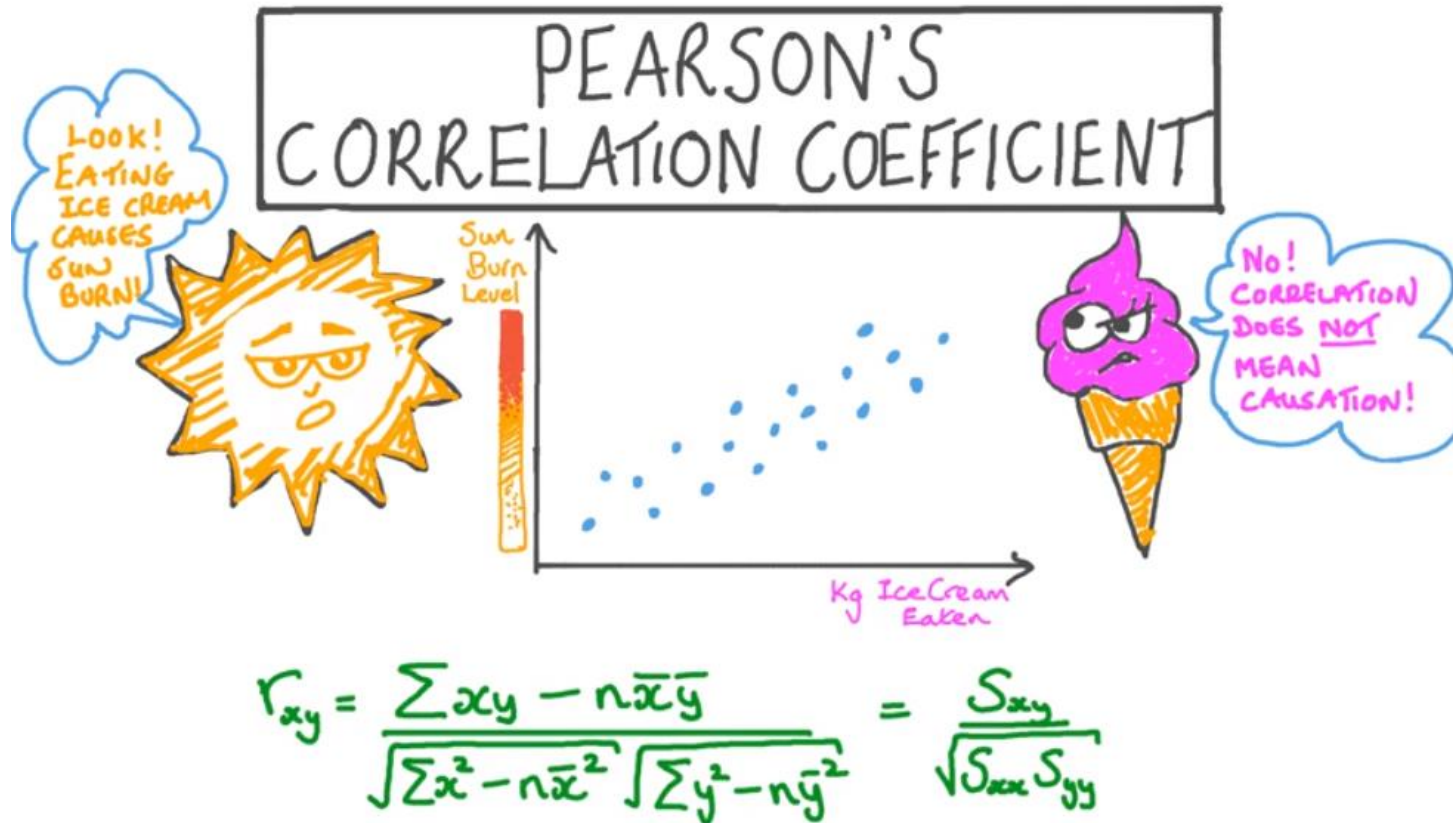


Table of Contents

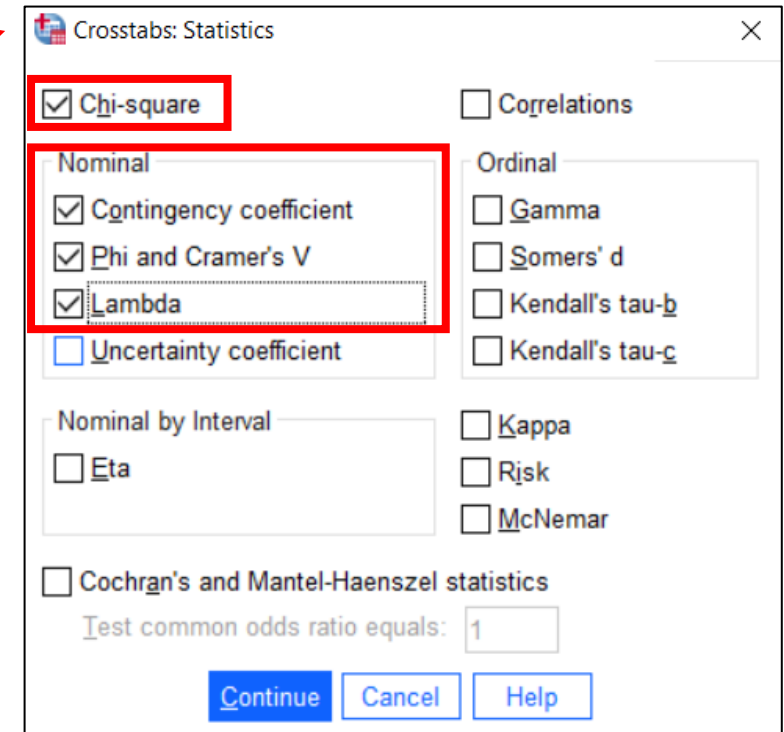
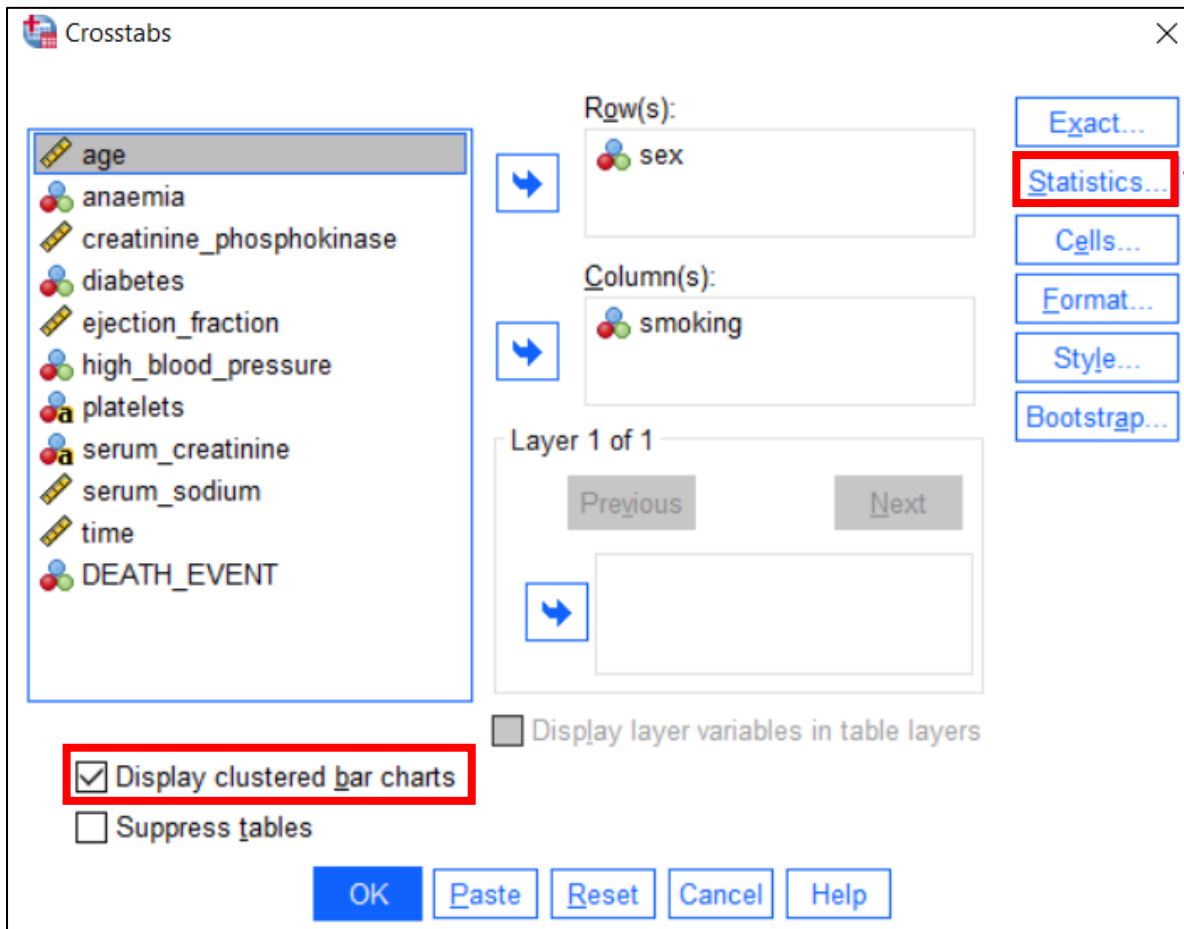
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Inferential Statistics

- **SPSS**

- **Chi-square test**

Analyse -> Descriptive -> Crosstabs

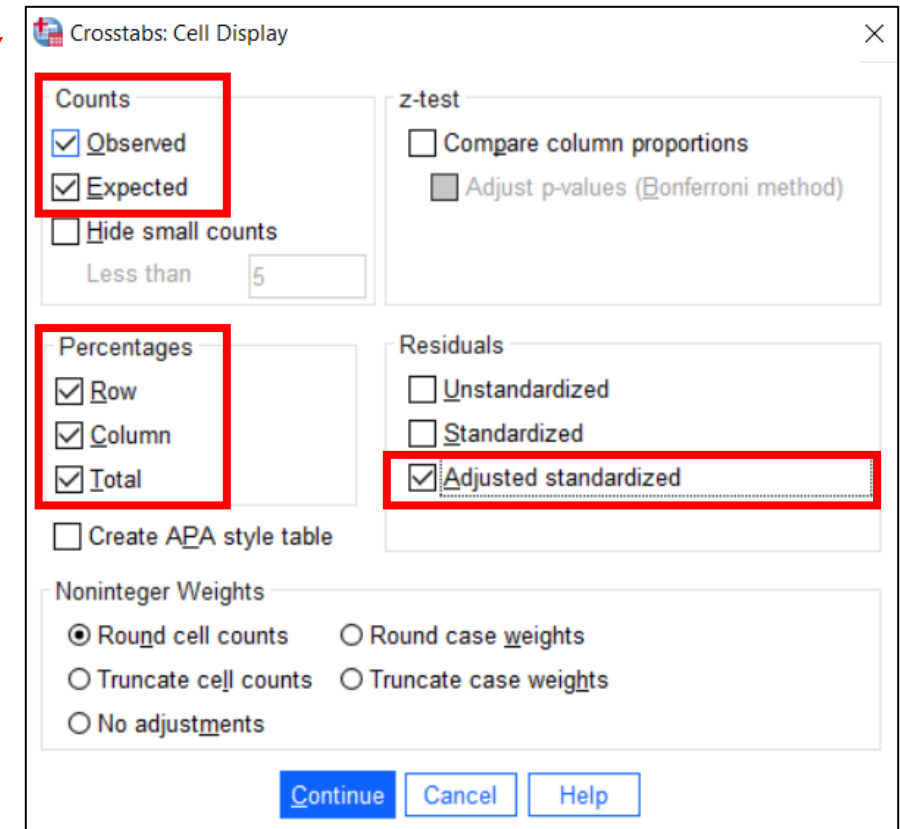
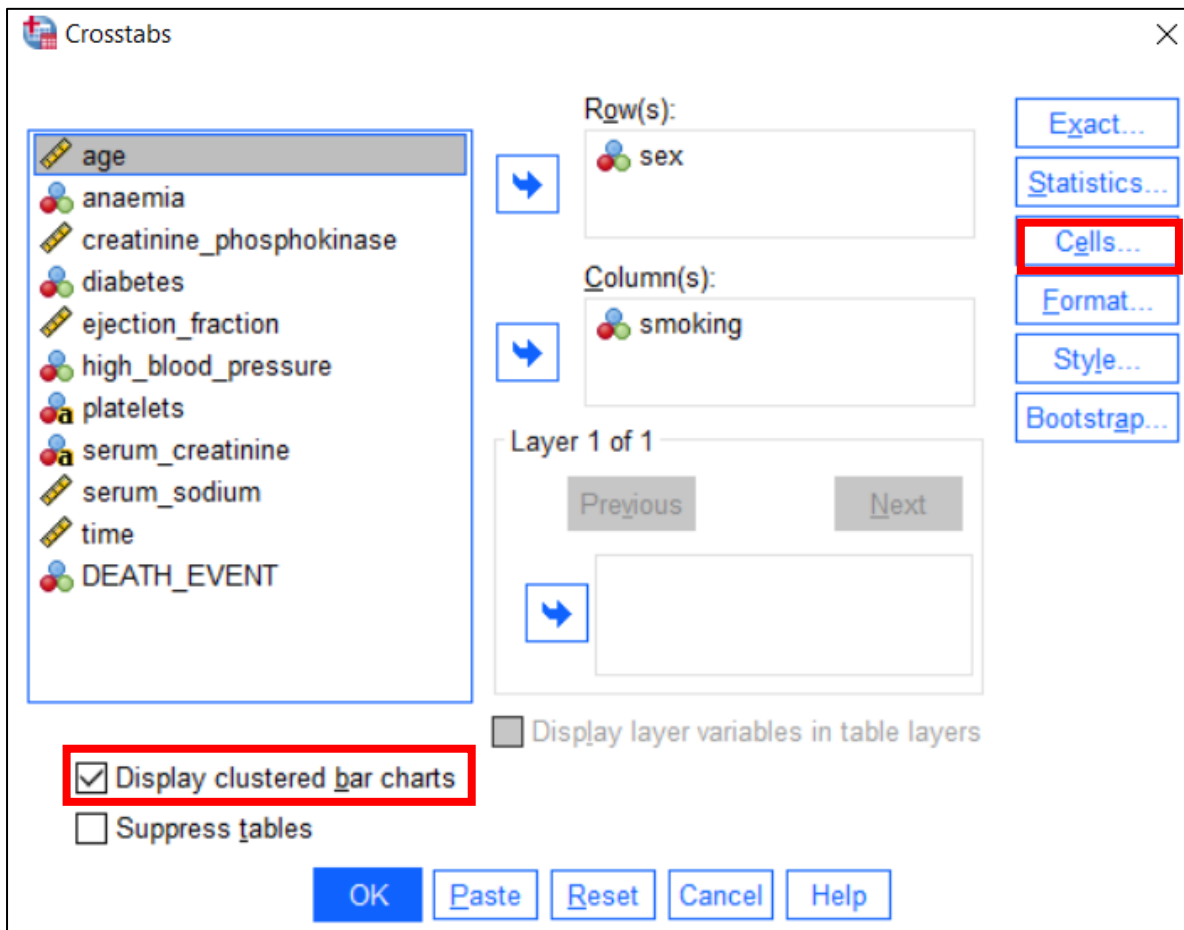


Inferential Statistics

- **SPSS**

- **Chi-square test**

Analyse -> Descriptive -> Crosstabs



Inferential Statistics

- SPSS
- Chi-square test

sex * smoking Crosstabulation

		smoking		Total	
		0	1		
sex	0	Count	101	4	105
		Expected Count	71,3	33,7	105,0
		% within sex	96,2%	3,8%	100,0%
		% within smoking	49,8%	4,2%	35,1%
		% of Total	33,8%	1,3%	35,1%
		Adjusted Residual	7,7	-7,7	
	1	Count	102	92	194
		Expected Count	131,7	62,3	194,0
		% within sex	52,6%	47,4%	100,0%
		% within smoking	50,2%	95,8%	64,9%
		% of Total	34,1%	30,8%	64,9%
		Adjusted Residual	-7,7	7,7	
Total	Count	203	96	299	
	Expected Count	203,0	96,0	299,0	
	% within sex	67,9%	32,1%	100,0%	
	% within smoking	100,0%	100,0%	100,0%	
	% of Total	67,9%	32,1%	100,0%	

[1,96; + ∞]: associação positiva
[- ∞ ; 1,96]: associação negativa

Inferential Statistics

- SPSS
- Chi-square test

Chi-Square Tests					
	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	59,447 ^a	1	<,001		
Continuity Correction ^b	57,463	1	<,001		
Likelihood Ratio	72,936	1	<,001		
Fisher's Exact Test				<,001	<,001
Linear-by-Linear Association	59,248	1	<,001		
N of Valid Cases	299				

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 33,71.

b. Computed only for a 2x2 table

sig < 0,05 » Rejeita-se a hipótese nula da independência das variáveis

Inferential Statistics

- **SPSS**
 - **Association Measures**

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	,446	<,001
	Cramer's V	,446	<,001
	Contingency Coefficient	,407	<,001
N of Valid Cases		299	

Inferential Statistics

- **SPSS**

- **Correlation Measures**

Analyse -> Correlate -> Bivariate



The image shows the 'Bivariate Correlations' dialog box in SPSS. On the left, a list of variables includes 'anaemia', 'diabetes', 'high_blood_pressure', 'sex', 'smoking', 'time', and 'DEATH_EVENT'. A blue arrow points from this list to the 'Variables:' box on the right, which contains 'ejection_fraction', 'age', 'creatinine_phosphokinase', and 'serum_sodium'. The 'serum_sodium' variable is highlighted. On the right side of the dialog, there are four buttons: 'Options...', 'Style...', 'Bootstrap...', and 'Confidence interval...'. The 'Correlation Coefficients' section is highlighted with a red rectangle and contains three checked options: 'Pearson', 'Kendall's tau-b', and 'Spearman'. The 'Test of Significance' section has 'Two-tailed' selected. At the bottom, there are three checked options: 'Flag significant correlations', 'Show only the lower triangle', and 'Show diagonal'. The bottom of the dialog features buttons for 'OK', 'Paste', 'Reset', 'Cancel', and 'Help'.

Bivariate Correlations

Variables:

- anaemia
- diabetes
- high_blood_pressure
- sex
- smoking
- time
- DEATH_EVENT

- ejection_fraction
- age
- creatinine_phosphokinase
- serum_sodium

Options...

Style...

Bootstrap...

Confidence interval...

Correlation Coefficients

☒ Pearson ☐ Kendall's tau-b ☒ Spearman

Test of Significance

☒ Two-tailed ☐ One-tailed

☒ Flag significant correlations ☐ Show only the lower triangle ☒ Show diagonal

OK Paste Reset Cancel Help

Inferential Statistics

- **SPSS**

- **Correlation Measures**

Analyse -> Correlate -> Bivariate

		Correlations			
		ejection_fraction	age	creatinine_phosphokinase	serum_sodium
ejection_fraction	Pearson Correlation	1	,060	-,044	,176**
	Sig. (2-tailed)		,302	,448	,002
	N	299	297	299	299
age	Pearson Correlation	,060	1	-,082	-,046
	Sig. (2-tailed)	,302		,160	,430
	N	297	297	297	297
creatinine_phosphokinase	Pearson Correlation	-,044	-,082	1	,060
	Sig. (2-tailed)	,448	,160		,305
	N	299	297	299	299
serum_sodium	Pearson Correlation	,176**	-,046	,060	1
	Sig. (2-tailed)	,002	,430	,305	
	N	299	297	299	299

** . Correlation is significant at the 0.01 level (2-tailed).

Inferential Statistics

- SPSS

- Correlation Measures

Nonparametric Correlations

Analyse -> Correlate -> Bivariate

Correlations			ejection_fraction	age	creatinine_phosphokinase	serum_sodium
<u>Spearman's rho</u>	ejection_fraction	Correlation Coefficient	1,000	,074	-,068	,162**
		Sig. (2-tailed)	.	,201	,242	,005
		N	299	297	299	299
	age	Correlation Coefficient	,074	1,000	-,092	-,101
		Sig. (2-tailed)	,201	.	,113	,083
		N	297	297	297	297
	creatinine_phosphokinase	Correlation Coefficient	-,068	-,092	1,000	,017
		Sig. (2-tailed)	,242	,113	.	,771
		N	299	297	299	299
	serum_sodium	Correlation Coefficient	,162**	-,101	,017	1,000
		Sig. (2-tailed)	,005	,083	,771	.
		N	299	297	299	299

** . Correlation is significant at the 0.01 level (2-tailed).

Bibliography

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